

# Part 15

# Database in Django

VERSION: 4.0.1

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# Database

*A database is a collection of data arranged in an efficient way. It ensures fast and secure storage, access and update of data. That is, the main concept of every type of database is both paper registers and MySQL. There are many types of databases out there. We will be narrowing them down to relational databases. Now, what's that? A relational database is a collection of tables having a number of columns and rows.*

Yes, there are databases that actually don't use columns and rows. So, whatever databases we have worked with on the web is a relational database. These databases can have relational tables or fields. We have used Foreign Key Field in our Django projects.

# SQL (Structured Query Language)

SQL (Structured Query Language) is a very popular language for databases. Almost all the popular databases use SQL. The databases like PostgreSQL, MySQL, SQLite, etc. are some of many examples. The databases mentioned above are relational databases. Yes, the majority of SQL databases are relational and are extensively used in the industry.

What's special about PostgreSQL, MySQL, etc. is that they have native support from Django. SQLite3 is the one that comes installed out of the box, although, you will have to install the software for other databases. Django ORM will handle the rest according to the backend engine used.

# Settings the Database

These are common settings you should know to connect database with Django project. This code exists in the **settings.py** file.

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.sqlite3',  
        'NAME': 'mydatabase',  
    }  
}
```

Now, these keys have different values for different databases. This configuration is for SQLite database.

# Note

Django comes with these four database backends natively. These databases will require their database software but the backend is in-house.

## PostgreSQL:

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.postgresql',  
        'NAME': 'mydatabase',  
    }  
}
```

# MySQL

## MySQL:

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.mysql',  
        'NAME': 'mydatabase',  
    }  
}
```

# Oracle

## Oracle:

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.oracle',  
        'NAME': 'mydatabase',  
    }  
}
```

These backends will be able to implement all the *latest features of Django*. They are developed and maintained by the **DSF (Django Software Foundation)**. You can use other backends as well. In that case, you have to import the full package path of that backend.

Thanks for Reading